Assignment 2 — Report

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Student: Ahmed Hannan Roll #: 22i-0943 Course: Agentic AI (Section C) Assignment #: 2

Executive summary

This assignment delivered a retrieval-augmented QA evaluation pipeline on the CRAG dev set. The implementation includes a robust loader, FAISS-based index creation, retrieval using sentence-transformers, multiple pipeline variants (basic RAG, multi-query RAG, rewrite-retrieve-read, HyDE, RAG-Fusion, CRAG and Graph-RAG stubs), an offline answer extractor, and an evaluation harness that computes normalized exact-match accuracy. The final evaluation (limit=100) reported per-pipeline accuracies saved in results_summary.json.

Implementation details

1. Data & index - A JSONL loader reads examples and extracts page snippets. - Unique snippets are embedded using the SentenceTransformer model and added to a FAISS index with L2 normalization to enable cosine similarity.

2. Retrieval - Queries are encoded with the same embedding model (normalized) and run against FAISS to retrieve top-k candidate passages. The retriever is fast and returns passages with similarity scores.

3. Generation / Answer extraction - The extractor is an offline component that uses spaCy NER (when available) to propose candidate answers (PERSON, ORG, GPE, DATE, CARDINAL etc.). Candidates are scored by passage similarity, match to question type (who/when/where/how many), multi-word preference, and token overlap. Robust regex fallbacks (e.g., 'directed by' and year extraction) are used when necessary.

Evaluation & results

The evaluation harness builds/loads the index, iterates examples, runs each pipeline, and computes a normalized exact-match score that tolerates punctuation differences and simple aliases. A typical run with limit=100 produced the following representative scores:

- basic_rag: 15/100 = 0.15 - multi_query_rag: 15/100 = 0.15 - rrr: 15/100 = 0.15 - rag_fusion: 13/100 = 0.13 - hyde: 15/100 = 0.15 - crag: 15/100 = 0.15 - graph_rag: 13/100 = 0.13

Limitations & future work

1. Generation: The current extractor is heuristic-based. Replacing it with a concise neural generator or an allowed local LLM can improve accuracy substantially.

2. Normalization: A curated alias map (canonicalization) for common entities (country names, company names, acronyms) will improve exact-match scoring.

3. Additional pipelines: Implementing a learned reranker or integrating passage-level QA models that extract exact spans rather than entities could further boost scores.

Submission checklist

Include the project folder with source, requirements.txt, the report PDF, and evaluation output. If the dataset is sensitive, provide instructions to rebuild the index rather than including the dataset.

End of report.